

Xavier S. Johnson

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EDUCATION

University of California Berkeley

• Master of Engineering | Mechanical Engineering – Product Design | GPA: 3.86 | GEM Fellow | Graduation Date: May 2025

University of Tennessee Knoxville

• Bachelors | Mechanical Engineering | Graduation Date: May 2024

WORK EXPERIENCE

Mechanical Engineering Intern

May 2024 – Aug 2024

ASML

San Diego, CA

- Designed mechanical assemblies using sheet metal design in NX12 in order to implement a heating stage in a Modular Vacuum chamber
- Prototyped and created sheet metal and plastic parts to integrate different electrical modules into test chambers and improve reliability testing
- Developed, performed calculations, and implemented a component-level test bench to test for adhesive delamination in the droplet generator
- Managed 4 different projects while collaborating with mechanical design and reliability engineers, vendors, and manufacturing

Product Design Engineer Intern

May 2023 – Aug 2023

Amazon Lab 126

Sunnyvale, CA

- Designed and prototyped a universal debug fixture ASM using CREO, incorporating pogo pins, PCBA, FPC flex cable, and springs
- Collaborated with tooling/mold engineers and contract manufacturers to ensure proper molding of plastic parts and seamless UDF assembly, applying DFM/DFA principles to optimize manufacturability and reduce production issues
- Conducted rapid 3-D prototyping and user study to ensure feasibility, user-friendliness, and proper functionality of components
- Created plastic and sheet metal parts to house electrical assemblies, collaborating cross-functionally with electrical, software, and tooling engineers, as well as CMs/External Vendors, to meet specified design requirements
- Performed FEA contact analysis in CREO Simulate for removable snaps, optimizing cantilever length, lead in/out angle, and width of snaps

Hardware Reliability Engineer Intern

May 2022 – Aug 2022

Amazon Lab 126

Sunnyvale, CA

- Researched and addressed in-field issues, collaborated with failure analysis, thermal engineers, and technicians to develop innovative solutions, and analyzed experiment data
- Developed component-level test and utilized HALT chamber for thermal aging to evaluate the thermal stability / leaching of different materials, enhancing part reliability and guiding supplier improvements
- Researched into different absorbent materials and glass/ceramics for test rig, ensuring test accuracy and performance
- Designed, prototyped, and created drawings for a test rig, drawing inspiration from various literature - in CREO

Mechanical Design Engineer Intern

May 2021 – Jan 2022

Siemens Healthineers

Knoxville, TN

- Conducted tolerance stack-up analysis (worst case) to ensure allowable misalignment of the belt drive in the Patient Handler System (PHS)
- Assisted development engineers in building and testing the Patient Handler System through life cycle, thermal imaging, and acoustic tests.
- Prototyped and designed sheet metal parts in Siemens NX 12 to improve safety/patient experience
- Performed machine design calculations to ensure components of PHS had allowable safety factors, allowable angle misalignments, and to find areas to improve such as using new materials, editing tolerances on drawings, and using new components
- Communicated with different suppliers/manufacturers to get quotes and ordered parts for engineers

Quality/Design Engineering Intern

August 2020-Jan 2021

Bosch Home Appliances

Oak Ridge, TN

- Designed different parts using principles of plastic part design in NX 12 and used 3-D printing to rapidly prototype parts
- Developed a Python script and GUI to interface with Home Connect API Simulators, enabling settings adjustments for refrigerators
- Designed parts for refrigerator assemblies and drafted new refrigerator assemblies in NX 12 and Teamcenter PDM
- Assisted quality engineers in developing a data analytical software program in Excel to reduce time sorting through failure symptom data and filter necessary information

UNDERGRADUATE RESEARCH

Undergraduate Research Assistant (Additive Manufacturing)

February 2022 – May 2022

Oak Ridge National Lab / University of Tennessee

Oak Ridge, TN

- Assisted in researching the material transitions of Carbon Fiber reinforced ABS to non-filled ABS made by a dual hopper extruder
- Prepared samples for analysis by using diamond saw, drying materials using heat oven, using chemical solvent and ultrasonication
- Performed strength testing on samples using a DMA

Undergraduate Research Assistant (Surgical Robots)

February 2021 – Sep 2021

University of Tennessee | Robotics Lab

Knoxville, TN

- Contributed to research on the use of robotic camera systems in minimally invasive surgeries (MIS), involving the design of components for a product utilizing a six DOF manipulator and other electrical elements using SOLIDWORKS
- Developed innovative housings to secure PCB, MC board, antenna and electronic modules in a fully enclosed environment and made connection to a separate force torque sensor and a actuator assembly easier
- Enhanced actuator assembly for ease of insertion of EPMS and micromotor

PROJECTS

Omni Direction Treadmill – Capstone Berkeley

Fall 2024 – Spring 2025

- Performed structural analysis on FEA frame in SOLIDWORKS simulate to ensure stability and verify material selection
- Resolved ball rotation issues via root cause analysis of enclosure tolerances, coatings, and assembly
- Conducted user studies and prototyped magnetism and friction based braking systems
- Redesigned frame from sand casting to CNC machined parts reducing cost & lead time while increasing manufacturability
- Machined t-slot framing and plates, ensuring precise alignment and fit for assembly

IEEE Robotics Club – SUMO BOT

Spring 2021

- Created a “sumo bot”, a robot that competes against other robots and attempts to knock them off a platform
- Created the robot for competition with a custom chassis, Mecanum wheels, an Arduino, driver boards, and a lever mechanism
- Created lever mechanism with wooden ramp to throw other robots off balance/undercut front wheels

Smart Speaker (Python Project)

Spring 2020

- Developed code for a custom speaker that focuses on appealing to the mental health of students and assists them in daily life.
- Created a speech recognition robot in Python that can tell the weather by utilizing an API, say different phrases based on input, give the time and date, and send an email including various resources that are tailored towards engineers

LEADERSHIP & INVOLVEMENT

Lead Ambassador – University of Tennessee Engineering Professional Practice

August 2023 – Spring 2024

- Produced instructional videos, conducted interviews with engineers, directed social media initiatives, resulting in a 500% surge in students engagement on social media
- Conducted engaging and informative presentations to students, effectively communicating program benefits, requirements, and opportunities
- Spearheaded a dynamic team of ambassadors, overseeing training, scheduling events, and improving student engagement

SKILLS

Software Skills: SolidWorks, Siemens NX, CREO, Fusion 360, Python, MATLAB, KiCAD, Nastran, SOLIDWORKS SIMULATE

Technical Skills: Sheet Metal, Plastic part design, 3D Printing, GD&T, Machining, DFM, DFA, Thermal & Static Simulations Tolerance Analysis, DOE, Manufacturing, FEA, HVM